Permutations with Indistinguishable Objects

- **Theorem**

The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, …, and n_k indistinguishable objects of type k, ($n=n_1+n_2+..+n_k$) is

$$\frac{n!}{n_1!\,n_2!\,\ldots n_k!}$$